

LESSON 326 (1.6 DUAL XC, 1.4 INSTRUMENT)

Lesson Objective

During this lesson, the student will plan and conduct a short IFR cross-country flight. During the flight, the student will become familiar with IFR departure, enroute, and arrival procedures.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

The purpose of this lesson is to demonstrate the proper sequence of tasks to complete on an IFR XC as well as expose students to the IFR ATC flow. Help them with radio communications and encourage the use of autopilot to reduce task saturation.

Opening and Closing an IFR Flight Plan

- File an IFR flight plan along with the student.
 - ForeFlight/EFB
 - 1-800-WX-BRIEF
 - 1800wxbrief.com
 - skyvector.com
- Remember to put your information, rather than the student's, as they do not yet possess an instrument rating.
- If arriving at a towered airport, the flight plan will automatically be closed.
- If arriving at a non-towered airport, ATC will ask whether you wish to close the flight plan in the

air or on the ground (frequency/phone number).

- Set a reminder to remember to cancel on the ground if desired.

Clearance Copying and Read Back

- As this may be the student's first time picking up a clearance, consider copying the IFR clearance along with the student and reading it back while the student listens.
- Emphasize the importance of proper avionics setup prior to taxi and takeoff; the flight plan should be loaded in the GPS, the departure heading and altitude should be bugged on the PFD, and the departure frequency and transponder code should be set prior to entering the movement area.

IFR Departure

- **Use of DP**
 - Review any published SIDs/ODPs and consider aircraft performance to ensure they can be complied with.
 - KSFB - Sanford usually assigns the Sanford One Departure, a simple radar vector-based SID. This should be briefed on the ground before the flight.
- **Use of radar**
 - Make sure the student correctly inputs assigned transponder codes as well as understands the ALT and IDENT functions.

Air Traffic Control Clearances

- Comply.

Emergency Operations

- **Loss of communication**
 - At some point during the flight, simulate a scenario where the student cannot hear ATC.
 - Ensure the student follows the correct steps of the IFR lost comms procedure:
 - Troubleshoot
 - Return to last assigned frequency
 - Check volume of COM1/COM2 and headset
 - Unplug/replug headset, try different jacks
 - Reset COM1/COM2 circuit breaker
 - Test Squelch function
 - Squawk 7600
 - Determine if VMC (Visual Meteorological Conditions) or IMC (Instrument Meteorological Conditions):
 - If VMC, land as soon as practicable.
 - If IMC, determine a route and altitude to fly to the destination:

Route - in order:

Assigned
Vectored
Expected
Filed

Altitude - highest of:

Minimum IFR Altitude
Expected
Assigned

- If the clearance limit is a fix from which an approach begins, begin the approach at EFC (or ETA, if none exists).
- If the clearance limit is not a fix from which an approach begins, leave clearance limit at EFC (or ETA, if none exists) and proceed to a fix from which an approach begins.
- **Loss of AD-AHRS**
 - At some point in the flight, simulate a failure of the AD-AHRS system.
 - Troubleshooting steps (**SECTION 3 - EMERGENCY PROCEDURES**):

Indication: Attitude and Heading Data removed and replaced with Red X's.

Standby Attitude GyroVERIFY ON and flag is pulled on gyro.

Maintain attitude control using standby gyro.

If time and conditions permit:

PFD Circuit BreakerPULL and RESET

If attitude and heading data is still invalid:

Maintain attitude control by using standby gyro.

Maintain heading control by utilizing magnetic compass and other directional indications (such as MFD, MAP/NAV page).

CAUTION

High current loads in the vicinity of the magnetic compass can influence its accuracy. Depending on the flight conditions, the pilot must reduce these loads as much as possible to insure accuracy. Tests have shown that air conditioner and pitot heat contribute to significant heading errors of the magnetic compass. These items should be turned OFF prior to comparing magnetic compass headings.

Land as soon as practical.

Precision and/or Non-Precision Approaches

- Encourage the student to complete the WIRE checklist as early as practical to reduce the workload upon arrival.
- Ensure the student completes all the steps of the WIRE checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.

- Radios:
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course.
- Enroute brief

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 327
- Assign Homework:
 - Plan another short XC (>50nm). If the current lesson was to a towered airport, consider going to a non-towered airport, or vice-versa.

Completion Standards

The student should:

1. Display an understanding of the procedures involved in cross-country planning, filing IFR flight plans, and obtaining IFR clearances.
2. Demonstrate understanding of the procedures used in the various emergency situations.

The lesson must include a landing at an airport with an instrument approach more than 50 NM from the departure airport.